Where to Predict in Retinal OCT? Anatomy-Guided Target Selection for I-JEPA

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QR CODE PENDING
paper / code link

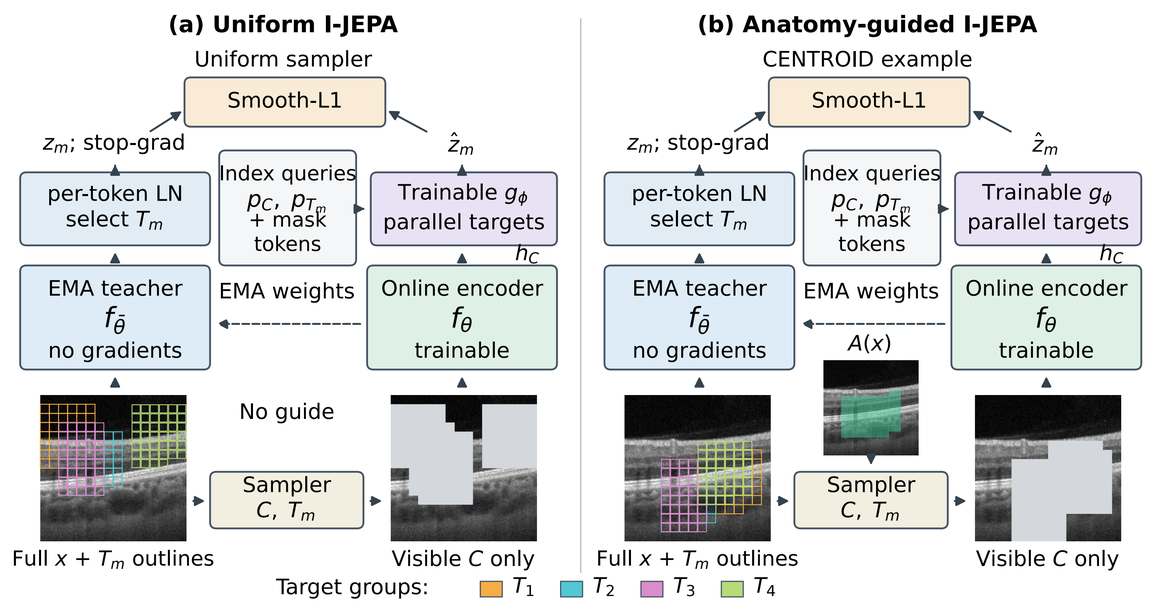
paper & code

Why target selection matters in OCT

- In retinal OCT the layered retina fills only a band of each B-scan; much of the rest is near-uniform background.
- I-JEPA learns by predicting the *latent features* of hidden target blocks from visible context, but places targets uniformly.

Question. Does steering I-JEPA targets toward retinal tissue give a better frozen encoder for glaucoma detection, with the same architecture and latent loss?

Same I-JEPA, different targets

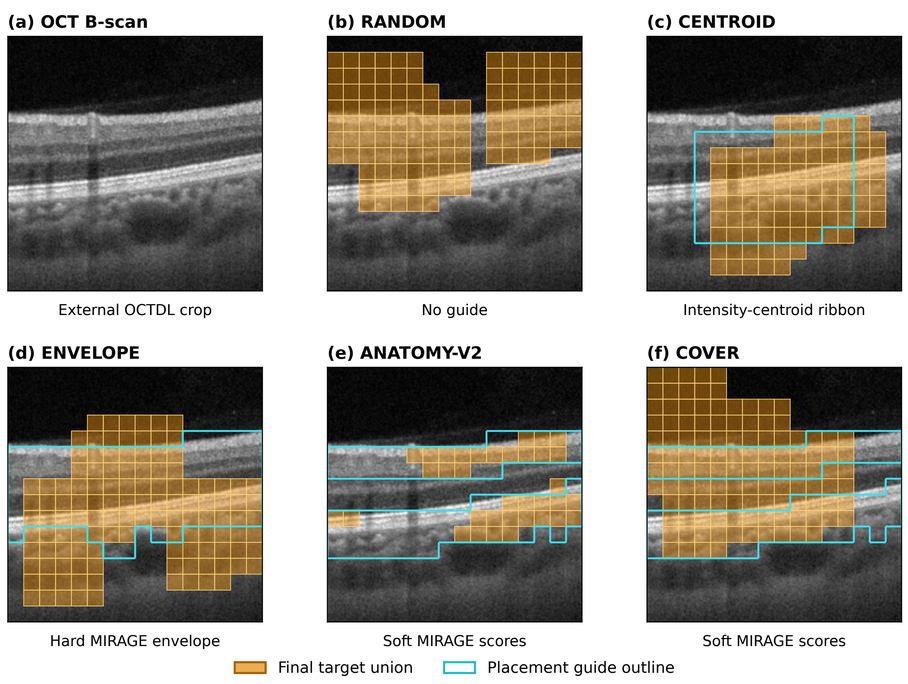


Evaluation: freeze EMA encoder → pool patches / B-scans → linear classifier

OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0); gray = omitted patches, not zero input.

Only the sampler of context C and targets T changes: $(C, T) \sim \Pi_s(\cdot \mid A_s(x))$ with the guide $A_s(x)$ of strategy s. Encoder, EMA teacher, predictor and Smooth-L1 loss are shared.

Five target-selection strategies



OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0), cropped. Illustration only; not FairVision.

strategy	guide $A_s(x)$ → target rule
RANDOM	none → uniform rectangles (stock I-JEPA)
ENVELOPE	hard MIRAGE retina envelope → rectangles sampled on it
CENTROID	intensity-centroid band, no segmenter → rectangles weighted by overlap
ANATOMY-V2	soft MIRAGE retinal scores → tissue-shaped components
COVER	soft MIRAGE retinal scores → greedy coverage rectangles

Original runs: tissue-directed rectangles help

FairVision glaucoma OCT; frozen EMA encoder, mean-pooled features, linear probe; test N = 3000 volumes (1466 glaucoma). All continuations start from one epoch-25 uniform I-JEPA checkpoint.

	epoch 50		epoch 75		epoch 100	
strategy	AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	0.8761	+0.0120	0.8803	+0.0080	0.8807	+0.0062
CENTROID	0.8740	+0.0099	0.8836	+0.0113	**0.8855**	+0.0109
ANATOMY-V2	0.8654	+0.0013	0.8612	−0.0111	—	—
COVER	0.8643	+0.0002	0.8639	−0.0084	0.8577	−0.0168

Δ: AUC minus RANDOM at the same epoch. Upper rows fp16 probes; lower rows fp32 probes against an fp32 RANDOM. ANATOMY-V2 stopped at epoch 75. One pretraining continuation per strategy: intervals reflect test-case sampling (10,000 paired bootstrap resamples), not retraining.

CENTROID, no segmenter: +0.0109 AUC at epoch 100 (95% CI [+0.0057, +0.0160]).
ENVELOPE: +0.0120 at epoch 50 (95% CI [+0.0068, +0.0173]).
More anatomy did not add gains: ANATOMY-V2 minus ENVELOPE −0.0107 at epoch 50 (95% CI [−0.0167, −0.0046]); COVER −0.0071 from its epoch-73 peak to epoch 100.

Which masks actually reach the model?

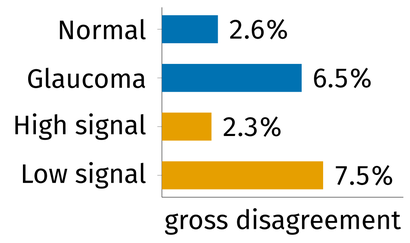
Sampler replay in each trained configuration: 6,400 training views, batch 64, 3 sampler seeds. Tissue is a MIRAGE-derived proxy, not ground truth.

strategy	tissue hidden	target purity	unique targets	loss slots	context (% grid)	AUC ep. 50
RANDOM	53.2%	30.9%	113.1	160.2	27.5 ± 0.1	0.8641
CENTROID	59.7%	38.2%	102.5	160.2	31.5 ± 0.4	0.8740
ENVELOPE	70.7%	42.1%	110.2	160.2	29.2 ± 0.5	0.8761
COVER	73.3%	40.1%	120.1	160.2	25.2 ± 0.4	0.8643
ANATOMY-V2	79.7%	96.9%	54.0	64.0	63.3 ± 0.4	0.8654

- Guided rectangles also leave more context visible than RANDOM: CENTROID +4.0 points (95% CI 3.6–4.4), ENVELOPE +1.6 (1.1–2.1).
- ANATOMY-V2 has the purest targets (96.9%) but 64.0 loss slots versus 160.2.

Target location, target size and visible context are coupled.

How reliable are the guides?



CENTROID band vs MIRAGE envelope: gross disagreement in 4.8% of B-scans (95% CI 3.7–5.9; all training volumes 5.3%).

Envelope plausibility flags: 3.8% (2.7–5.0).

MIRAGE guide: GOALS foreground Dice 0.925, in-domain only.

Agreement between two automatic guides, not anatomical accuracy (FairVision has no layer labels). Bars: unweighted rates in a stratified sample of 300 volumes (3,000 B-scans); high/low signal: top/bottom tertile of a signal-to-noise proxy.

Takeaways

- Tissue-directed targets improved frozen glaucoma probes in the original runs, even with a segmentation-free intensity centroid.
- More anatomical specificity (tissue shapes, coverage) added no further gain.
- Target placement, target size and visible context move together; matched-budget comparisons come next.

Registered replication: results pending (filled after the Oct 22 results freeze)

Repeated continuations Epoch-50 test AUC; originals re-probed, new runs from the epoch-25 checkpoint. Mean and Δ (vs RANDOM, same seed): registered seeds.

strategy	original	new seeds	mean	mean Δ
RANDOM	TBD	TBD	TBD	—
ENVELOPE	TBD	TBD	TBD	TBD
CENTROID	TBD	TBD	TBD	TBD

Matched-budget control RANDOM-CB: uniform placement with CENTROID's per-image budgets; seed A, one run per arm.

contrast	Δ AUC
CENTROID − RANDOM-CB (placement)	TBD
RANDOM-CB − RANDOM (budget)	TBD

Pooling Maxima within or across B-scans; mean Δ vs RANDOM (descriptive).

max	within	across
ENVELOPE	TBD	TBD
CENTROID	TBD	TBD

FIGURE PENDING
spread across runs
auto/fig_cr_seed_spread.pdf

Error bars: s.d. over probe-head seeds. Open markers: descriptive runs (late or incomplete).

RESULTS PENDING Labels (consistent direction, within run-to-run variation or reversed), cells and figure fill in from auto/cr_results.tex.

I-JEPA: Assran et al., CVPR 2023. MIRAGE: Morano et al., npj Digit. Med. 2025. FairVision: Luo et al., 2024. OCT example in both OCT figures: Kulyabin et al., OCTDL, Fig. 1 (arXiv:2312.08255v3), adapted (cropped, overlays added) under CC BY 4.0 (creativecommons.org/licenses/by/4.0); illustration only. No FairVision images are shown.